

SIMULATING COMPOSITION CHANGES IN BIOLOGICAL MEMBRANES WITH POP-MD

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ABSTRACT

The majority of molecular dynamics (MD) simulations of biological systems are performed under thermodynamic equilibrium, whereas life is inherently a non-equilibrium phenomenon. Many biological processes occur on timescales far beyond the reach of current MD simulations, which are typically limited to the microsecond-to-millisecond range [1]. To investigate how biological membranes respond to dynamical changes in composition, we introduce POP-MD, a simulation protocol that enables continuous modification of the number and identity of particles within user-defined regions of an MD simulation. By adding, removing, or transforming molecules during a simulation, POP-MD allows to probe, during nanosecond to microsecond time scales, the response of membrane systems to changes in molecular composition that experimentally take place in milliseconds to seconds time scales. As an example, POP-MD reveals that asymmetric phospholipid synthesis is necessary to drive lipid droplet budding, bypassing the rate-limiting metabolic reactions responsible for lipid synthesis [2]. We further demonstrate its versatility by dynamically constructing starting configurations, emulating lipid flip-flop, accelerating lipid mixing and demixing, and simulating vesicle growth by asymmetric phospholipid synthesis. Although POP-MD is not a rigorous framework for non-equilibrium dynamics, the vastly accelerated kinetics enables the simulation of processes that remain inaccessible to conventional MD and provides a practical framework for investigating composition-driven biological phenomena at molecular resolution, paving the way to simulations of organelle biogenesis.

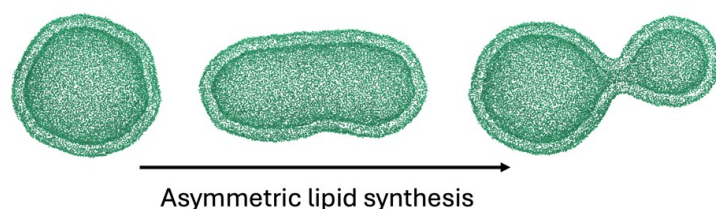


Fig. Simulating vesicle growth with POP-MD by asymmetric synthesis of phospholipids in the two leaflets.

REFERENCES

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